The Whole Thing, In Plain Words

No jargon, plus a glossary of every term used elsewhere

Dheirav

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The whole thing, in plain words

No jargon in this document. Every technical term used in the other documents is defined at the end.

The situation

A bank uses a computer program to decide who gets a mortgage. The program looks at income, loan size, property value and so on, and says yes or no.

Someone checks the results and finds a problem: the program approves **79%** of one group of applicants and only **57%** of another. Same kind of applications, very different outcomes.

That is the thing people call "algorithmic bias", and there are well-known tools for fixing it. You tell the program: *approve both groups at the same rate.*

The catch, and it is a known one

There are two completely different ways for the program to obey that instruction.

Way one: approve more people from the group that was being turned down.

Way two: turn down more people from the group that was getting through.

Both make the rates equal. Both count as "fixed". **And the score that certifies the fix cannot tell them apart** — it only measures whether the two rates match, not how they came to match.

Way one helps people. Way two helps nobody — it just spreads the disadvantage around. There is a name for it in the ethics literature: *levelling down.*

What was already known

A well-known 2023 paper from Oxford showed that these tools mostly do **way two**. They put it in their title: levelling down happens "by default".

And that the score cannot tell way one from way two was known too. A 2023 paper on fairness across combinations of groups says as much in plain words — this "often goes unnoticed in the overall performance of the model" — and a separate 2023 auditing tool exists because a fairness score does not say how many people were affected or in which direction. So the catch above is where this work starts, not something it found.

That paper is also the closest competitor — because it not only identified the problem, it proposed the fix. This project originally believed it had invented that fix independently. It had not. Found by reading their paper properly, and corrected.

The way of measuring it is not mine either. A 2023 paper set out how to audit what one of these fixes does to individuals: how many people it moves, which way it moves them, and what happens to the overall approval rate. Those are the three things counted here. What is mine is what the counts say once you run them on 26 runs across 15 populations instead of one — the direction turns out to depend on the data, which their framework was never used to ask.

And there is a second one. A March 2026 paper proves mathematically that when the program cannot see which group someone belongs to — which is the situation studied here — the effect **can go either way** depending on the data. So the *idea* that it is not always harmful was published, in theory, five months before this work reached it independently.

That paper has no experiments in it at all. Not one dataset. It proves the possibility and gives conditions written in terms of the shape of the program's internal scores.

What this work found

It is not "by default". It depends on how generous the system is to begin with.

Count what fraction of *everyone* gets approved, across both groups. Call that the approval rate of the system as a whole.

- If the system approves **fewer than about half** of everyone, forcing equality takes approvals **away**. Way two. The harmful one.
- If it approves **more than about half**, forcing equality **hands approvals out**. Way one. The good one.
- The switch sits close to an approval rate of 0.54, and everywhere it has been measured carefully it lands between roughly 0.43 and 0.58.

Same tool. Same instruction. Opposite effect on real people, depending on nothing more than how generous the system already was.

How it was proven, and why the proof is trustworthy

The obvious worry is that two datasets differ in a hundred ways, so you can never tell which difference caused what.

So the test used **one** dataset and changed exactly **one** thing.

The dataset predicts whether a person earns more than \{}$50,000. That \{}$50,000 is an arbitrary number someone picked. Move it to \{}$70,000 and fewer people qualify. Move it to \{}$20,000 and most people do.

Same people. Same information about them. Same groups. **Only the finish line moved.**

And the effect flipped, exactly as described. There is automated test code that checks the people, the information and the groups really are identical between runs, so this is not a matter of trust.

It then held up in:

- a second population,
- a second way of dividing people into groups,

- **real mortgage data**, where there is no arbitrary cutoff at all — the answer is a real lender's real decision,
- and **fourteen further runs — four new populations, each measured at several income cutoffs — carried out *after* the prediction was written down**, so they could not have influenced it. Four is the number that matters: runs on the same population share their people and differ only in where the cutoff was put.

What happened when I tried hard to break it

Since the last version I attacked this finding four ways, on five populations each instead of one. Two of the attacks it survived. Two found real limits, and both are reported here as limits rather than tucked away.

I tried	Asking	What happened
a completely different kind of model	is this just true of simple models?	**held, 5 out of 5**
making the fairness rule much stricter, and much looser	is it just one setting?	**held, 4 out of 5**
reaching the same approval rate a different way	is it the approval rate, or the difficulty?	**held, 4 out of 5**
a different definition of fairness	does it hold for any fairness rule?	**it failed**

The third one matters most. The original evidence changed the approval rate by changing what counted as success — earning over \$20,000 instead of over \$100,000. But those are not the same task; one is much harder to predict than the other. So the old evidence could never separate *"the approval rate is what drives this"* from *"the difficulty is what drives this."*

The new test changes nothing about the task at all. Same people, same information, same predictions — only the cut-off the model uses to say yes. **The approval rate is what matters.** And it also cleared up a puzzle I could not explain before: the switch point sits in a different place for mortgages than for income, and that turns out to be a real difference between the two worlds rather than a mistake in how I built the earlier evidence.

The fourth one failed, and I am not going to dress it up. There is a second common definition of fairness, concerned with error rates rather than with how many people get approved. Under that definition the relationship almost disappears, and adding more populations made it weaker rather than stronger. So the finding is narrower than I first claimed: it is about fairness rules that control **how many people are approved**, not about fairness rules generally.

One population gave no answer at all. In Connecticut almost everybody is above the switch point already, so the fairness rule barely changes anything and there is nothing to measure. That turned up a flaw in my own method — my original test demanded that the relationship be strong, but never demanded that anything actually *move*. I have added that requirement and re-checked every earlier result against it. Three go from "failed" to "no answer"; **none of them had ever passed by accident**, so nothing I claimed turns out to be wrong. But the hole was mine, and it was found by a population rather than by me.

Two things I learned since the last version

Where the switch sits is much more predictable than I thought. Measuring it properly — more points, and throwing away any test where the model had become worse than useless — four completely different populations agree:

population	field	switch point
COMPAS	criminal justice	0.511
South Carolina	income	0.530
Oregon	income	0.558
Dutch census 2001	jobs	0.576

Three fields, two countries, four separate data sources — all between **0.51 and 0.58**. So the advice is now "expect about 0.54 and check", instead of "you'll have to work yours out from scratch".

And the finding belongs to one specific situation — which the theory predicted. There are two ways to make a system fairer. You can change how it is *trained*, which is what I do. Or you can leave the training alone and *adjust the decisions afterwards* — but that requires knowing each person's race or sex at the moment you decide, which is illegal in most places for exactly that reason.

I tried the second way across seventeen populations. **My finding disappeared completely.**

That sounds bad, and it is the opposite. The theory paper says these two situations are fundamentally different: when you *can* see the protected attribute, the outcome is **guaranteed** — the disadvantaged group always gains and the advantaged always loses. When you *cannot*, it could go either way, which is when a rule for telling which is worth having.

So there was nothing for my rule to predict in the second case. And the theory's prediction for that case held in **27 out of 27** of my populations, with no exceptions — nine of those predicted in advance, on populations that had never been measured that way before.

My claim is now smaller and firmer: it is about systems that **cannot** see the protected attribute when they decide — which is most of them, because looking at race or sex at the moment of the decision is illegal in most regulated settings.

How this sits with the theory

Three things came out of checking this work against that 2026 paper:

1. **Their conditions never actually hold.** Applied to all 26 real runs here, the mathematical conditions in their theorem are satisfied **zero times**. The quantity they are written over blows up on real data. So their result, as stated, cannot be used on anything.
2. **Their direction is still right.** Loosened into a comparison rather than a strict requirement, it predicts correctly in 24 of 26 cases.
3. **And it matches the approval-rate rule almost exactly** — the two agree on 25 of 26 datasets, and the numbers behind them correlate at **0.93**.

So: their maths explains *why* it happens. This work shows *how you can tell in advance* — using a number any organisation already has, rather than one requiring the model's internals. Two independent routes to the same place, which is better evidence than either alone.

The strongest test: predictions written down before the data existed

Everything above was measured by running experiments and then looking at the results, and the honest problem with that is well known: with enough looking, a pattern can always be found. So the rule was put through the one kind of test that looking cannot pass.

The idea is simple. Write the rule down. Name the datasets it will be tested on — ones nobody has ever measured. Write down the score it must reach, and write down what a lucky guesser would score, because the rule has to beat luck, not just do well. Save all of that in the project's history, where the timestamps cannot be faked. *Then* run the experiments.

The first attempt failed, and the failure taught the most important lesson in the project. Two hours before the test was locked in, I had added a clever extra clause to the rule, based on four datasets I had already seen. With that clause, the sealed test scored 4 out of 8 — no better than guessing. Without it, the same test would have scored 7 out of 8. A refinement that looked completely convincing on the data I had was exactly what ruined the prediction on data I had not seen.

So the clause was thrown away, and the plain rule — below the switch point means take away, above means hand out — was sealed against **ten more states nobody had touched**, with the pass mark set at 9 out of 10 and the requirement to beat the best possible lucky guesser, which scored 6.

It scored 9 out of 10. The one miss is counted as a miss, no excuses — though it is worth knowing that the missed state's true effect was about four hundredths of a percent, flipping sign between random re-runs, which means there may have been nothing there to predict in either direction.

One more thing about reading this project's record. Seven of these written-down-in-advance tests were run in total, and five failed. That is not a bad score, because they were not seven attempts at the same thing: each failure tested a *bigger* claim than the one that survives, and each one drew a boundary — the rule does not extend to a second definition of fairness, does not survive into the see-the-attribute world, does not come with an explanation of *why* it works. The five failures are the map of where the claim ends. The two passes show that inside that map, the prediction works — and every failure is printed in the paper, uncorrected, which is what makes the passes worth believing.

Why it matters

Loans. Job applications. University admissions. These are the places these tools actually get used — and I checked where each one sits.

what is being decided	share who get a "yes"	which way the fix goes
Sifting job applications	**2–3%**	takes opportunities away
Getting into a top university	4–10%	takes opportunities away
Getting into university generally	66–73%	hands opportunities out
Getting a mortgage	**~84%**	hands opportunities out

They are at **opposite ends**. So:

Two organisations can use exactly the same fairness tool, in good faith, and get opposite effects on how many people get helped. That the fairness report looks identical either way is not my observation — it is why Maheshwari et al. and Ferry et al. wrote their papers. What is new is that the *direction* differs between them, and that you can tell in advance which one you are about to get.

A bank fixing its mortgage model will end up approving more people. A company fixing its CV-screening will end up interviewing fewer. Same tool. Same score. Opposite result.

That is why knowing which side you are on matters, and why it is worth being able to check.

What could not be worked out

Why the flip happens.

There was an attempt: derive it from theory, write the prediction down in advance, then test it on new data. It passed the test that was set for it — and then a much stupider rule ("more than half get approved? then it's the good kind") did *better*.

So the honest position is: we know **when** it flips. We do not know **why**. That failure is written into the paper rather than left out.

The one number that is worth remembering

About 0.54 — call it "roughly half". Below that approval rate, these fairness tools take opportunities away. Above it, they hand them out. (An earlier version of this document said 0.3; measuring the switch point properly, on more data with better checks, moved it, and that correction is part of the record rather than papered over.)

And you can check which side you are on before building anything, using a number you already have.

Glossary

Terms used in the other documents in this folder.

Algorithmic bias. A program producing systematically different outcomes for different groups of people, usually because it learned from data in which that difference already existed.

Protected attribute. The characteristic the fairness rule is about — sex, race, and so on. In this work the programs are never allowed to see it directly.

Proxy. A piece of information that stands in for the protected attribute without being it. If almost everybody labelled "husband" is male, then that label is a proxy for sex, and removing "sex" from the data does not remove the information.

Demographic parity. The rule "approve both groups at the same rate". The main fairness rule studied here.

Constraint. The instruction given to the program. "Approve both groups at the same rate" is a constraint.

Mitigation. Any method for reducing bias. This work studies methods that change the program's *instructions* rather than editing the data.

Selection rate. The fraction of everyone who gets the good outcome. If a bank approves 8,000 of 10,000 applications, the selection rate is 0.8. **This is the central quantity in this work.**

Base rate. How often the good outcome genuinely occurs in a group — as opposed to how often the program predicts it.

Levelling down. Making two groups equal by making the better-off group worse, rather than the worse-off group better.

Levelling up. The opposite, and the desirable version.

"The pie." Informal shorthand for the total number of approvals. Levelling down shrinks the pie; levelling up grows it.

Exchange rate. How many approvals were destroyed for each one created. Below 1.0 means more were created than destroyed — good. On one dataset here it reached 46.7, meaning 46 people lost an approval for every one who gained one.

Baseline. The ordinary program, before any fairness fix, used as the comparison point.

Seed. A random number that decides how data is split for testing. Everything here is run five times with five different seeds, so that a result cannot be an accident of one split.

Held out. Data deliberately not looked at until after a prediction is written down, so the prediction cannot have been shaped by it.

Pre-registration. Writing down what you expect to find, and what would count as being wrong, *before* running the experiment. The commit history proves the order.

Feature attribution / SHAP. A tool that claims to show which pieces of information a program relied on. This work finds it can move dramatically without the program's actual behaviour changing much.

Intersectional. Looking at combinations — not just "women" and "Black applicants" but "Black women" specifically. A fix can look fine for each group separately while a combination is treated badly.

Correlation (r). A number from -1 to $+1$ for how strongly two things move together. $+0.9$ is a strong positive relationship; 0 is none.

Partial correlation. The same, but with a third factor held fixed, to check the relationship is not really caused by that third thing.

Adult / ACS / HMDA. The three data sources. *Adult* is a 1994 US census extract, the standard benchmark. *ACS* is its modern replacement, available per state. *HMDA* is a public record of real US mortgage applications and the lender's actual decision.

Population. One dataset — a state, a cohort, a census — counted once no matter how many experiments run on it. Thirty-nine have been measured at the time of writing, with ten more finishing overnight.

Sealed test. A prediction written down and saved in the project's history before the data it predicts about existed, together with its pass mark and the lucky-guesser score it must beat. The strongest form of evidence in this project, and the only kind that cannot be produced by hunting through results until something looks good.